

9.2 RLHF

SFT, then a reward model, then a policy that climbs the reward without wandering too far.

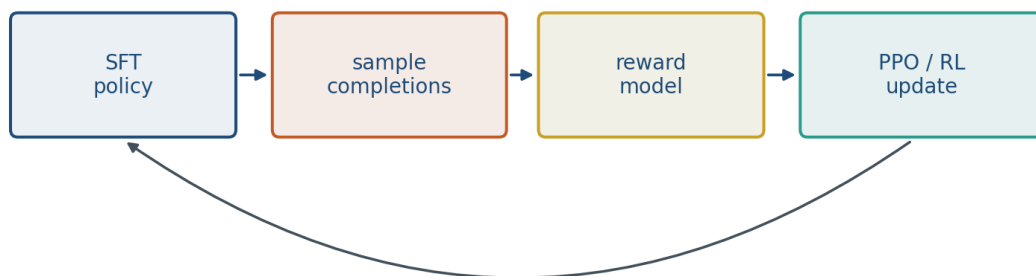
These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

RLHF (InstructGPT / ChatGPT-style) is a three-stage stack: SFT, a reward model, then a **policy** optimized with RL (usually **PPO**) so sampled replies score high under the RM without wandering too far from SFT.

1. The loop

1. **SFT** on instruction pairs (note **8.1**). Call this π_{SFT} .
2. **RM** r_ϕ on chosen/rejected pairs (note **9.1**).
3. **PPO**: sample $y \sim \pi_\theta(\cdot \mid x)$, score with r_ϕ , update π_θ .

RLHF: climb the reward, stay near the SFT policy (KL penalty)



PPO is on-policy: you need fresh rollouts from the current π_θ . That is expensive (generate, reward, critic). The **critic** (value head) estimates advantages so the policy gradient has lower variance. You do not need to derive PPO in this course; you need the picture and the KL term.

2. The KL penalty

Without a constraint, π_θ **hacks** r_ϕ : long, flattering, or degenerate text that the RM likes. The usual objective looks like

$$\mathbb{E}_{x, y \sim \pi_\theta} [r_\phi(x, y) - \beta \text{KL}(\pi_\theta(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x))],$$

with $\pi_{\text{ref}} = \pi_{\text{SFT}}$ frozen. β is the knob: large β stays near SFT; small β chases reward. In code this is often a per-token penalty $\beta \log \pi_\theta - \beta \log \pi_{\text{ref}}$.

3. Why people look past PPO

It works at company scale and is finicky in a seminar: sampling, KL, reward scale, and a moving policy.

DPO (next note) skips the RM and the loop. RLHF is still the conceptual parent: preferences $\rightarrow$ a scalar of “better” $\rightarrow$ a policy that maximizes it with a leash.

For a project, running full PPO on an LLM is usually out of scope. Best-of- n with a tiny RM, or DPO on a small classifier, is enough to show you understood the stack.

4. Practice

1. Which of the three stages needs *samples from the current policy*, not a fixed dataset?
2. If $\beta = 0$, what failure mode should you expect?
3. Why is π_{ref} the SFT model rather than the raw pretrained LM?